NARAYANA ENGINEERING COLLEGE :: GUDUR

DEPARTMENT OF ECE

CourseDetails							
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CourseTitle	:	**EMTL**	**CourseCode**	: **23EC2005**	**Credits**	:	**3**
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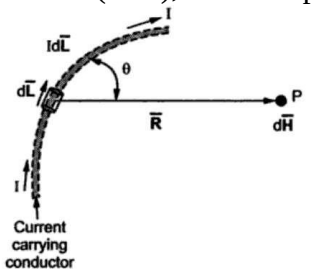
Regulation	:	**NECR-23**	**Faculty**	**:Dr. B. Malakonda Reddy**			

Question Bank /Topic Learning Outcomes

(Table format as given below is suitable for Theory, MCQ, Lab, and Tutorial. Columns can be added or deleted. No need to take printouts)

1	Find an expression for magnetic field intensity using Biot – Savart's Law. ANS: Consider a conductor carrying a direct current I and a steady magnetic field produced around it. Consider very small part of current carrying conductor i.e., differential current element(IdL), which is product of differential length dL and current I.  The point P is at a distance R from differential current element. The θ is angle between differential current element (IdL) and the line joining point P to differential current element. The Biot Savart Law allows us to find the differential magnetic field intensity dH , produced at point P, due to differential

current element (IdL)

The Biot Savart Law States that, The magnetic field intensity dH , produced at point P, due to differential current element (IdL) is

i. Proportional to the product of current (I) and differential element (dL)
ii. Proportional to the sine of the angle between differential current element (IdL) and the line joining point P to differential current element
iii. and inversely proportional to square of the distance R between point P and the element

Mathematically, the Biot Savart Law can be stated as,

$$dH \propto \frac{I\,dL\sin\theta}{R^2}$$

$$dH = \frac{k\,I\,dL\sin\theta}{R^2}$$

Where k = Constant of Proportionality

In SI Units $k = \frac{1}{4\pi}$

$$dH = \frac{I\,dL\sin\theta}{4\pi R^2}$$

<u>Vector form of Biot Savart Law</u>

Let dL = Magnitude of vector length $d\bar{L}$ and

$\bar{a}_R$ = Unit vector in the direction **from differential current element to point P**

Then from rule of cross product,

$$d\bar{L}\times\bar{a}_R = dL\,|\bar{a}_R|\sin\theta = dL\sin\theta$$

from above expression

$$d\bar{H} = \frac{I\,d\bar{L}\times\bar{a}_R}{4\pi R^2}\ \text{A/m}$$

But $\bar{a}_R = \frac{\bar{R}}{|\bar{R}|} = \frac{\bar{R}}{R}$

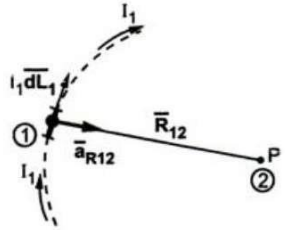
Hence

$$d\bar{H} = \frac{I\,d\bar{L} \times \bar{R}}{4\pi R^3} \text{ A/m}$$

This is mathematical form of Biot Savart Law in vector form.
The total magnetic field intensity H is obtained as

$$\bar{H} = \oint \frac{I\,d\bar{L} \times \bar{a}_R}{4\pi R^2}$$

The closed line integral is required to ensure that all current elements are considered. This is because current can flow only in the closed path, provided by the closed circuit.
Consider, if current element is at point 1 and point P at point 2,



then

$$d\bar{H}_2 = \frac{I_1\,d\bar{L}_1 \times \bar{a}_{R12}}{4\pi R_{12}^2} \text{ A/m}$$

where I_I = Current flowing through dL_I at point I

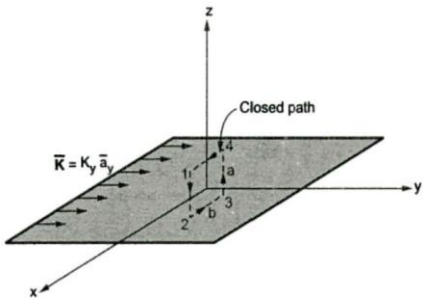
$d\bar{L}_1$ = Differential Length vector at point I

$\bar{a}_{R_{12}}$ = Unit vector in the direction from element at pointI to the point P at point 2

$$\bar{a}_{R12} = \frac{\bar{R}_{12}}{|\bar{R}_{12}|} = \frac{\bar{R}_{12}}{R_{12}}$$

$$\bar{H}_2 = \oint \frac{I_1\,d\bar{L}_1 \times \bar{a}_{R12}}{4\pi R_{12}^2} \text{ A/m}$$

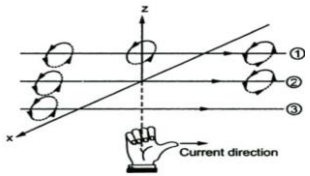
This is called integral form of Biot Savart Law

2	**Derive Maxwell's equations for Magnetostatic Fields.** ANS: Maxwell's equations •**Maxwell's equations** for magnetostatics: $\nabla \cdot B = 0$ $\nabla \times H = J$ Where; J = current density H = magnetic field intensity B = magnetic flux density •Relationship between B and H: $\mathbf{B} = \mu \mathbf{H}$ unit: Tesla or Weber/m^2 Where: μ = magnetic permeability
3	Derive an equation for magnetic field intensity due to Infinite sheet with uniform current density placed in z = 0 plane. ANS: Consider an infinite sheet current in z = 0 plane  The surface current density is $\bar{K}$ As the current is flowing in positive y-direction $\bar{K} = K_y \bar{a}_y$

Consider closed (Amperian) Path 1-2-3-4-1. Its width is ‘b’ and height is ‘a’. It is perpendicular to the direction of current hence in xz-plane. Current enclosed by the closed path 1-2-3-4-1which is flowing across the normal width ‘b’ is

$$I_{enc} = K_y b$$

According to right hand thumb rule



As current in y-direction, the magnetic field is normal to the current direction i.e., in xz –plane. The magnetic field is in xz -plane, at the same time it is tangential to the xy-plane, from this, only x-components will consider, by cancelling z-components.

So $\overline{H}$ has only component in x-direction

$\overline{\mathbf{H}} = \mathbf{H_x}\ \overline{\mathbf{a}}_\mathbf{x}$ ... for z > 0

$= -\mathbf{H_x}\ \overline{\mathbf{a}}_\mathbf{x}$ ... for z < 0

To Apply Ampere’s Circuital Law

Let us find $\overline{H}\cdot\overline{dL}$ in closed

$$(\overline{H}.d\overline{L})_{1\text{-}2} = -H_x\overline{a}_x.dz(-\overline{a}_z) = H_x dz(\overline{a}_x.\overline{a}_z) = 0$$

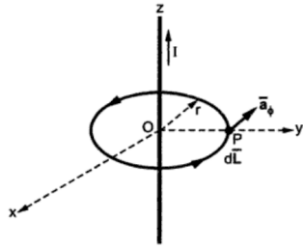
$$(\overline{H}.d\overline{L})_{2\text{-}3} = -H_x\overline{a}_x.dx(-\overline{a}_x) = H_x dx(\overline{a}_x.\overline{a}_x) = H_x dx$$

$$(\overline{H}.d\overline{L})_{3\text{-}4} = H_x\overline{a}_x.dz\overline{a}_z = H_x dz(\overline{a}_x.\overline{a}_z) = 0$$

$$(\overline{H}.d\overline{L})_{4\text{-}1} = H_x\overline{a}_x.dx\overline{a}_x = H_x dx(\overline{a}_x.\overline{a}_x) = H_x dx$$

Let us take integral of $\overline{H}\cdot\overline{dL}$ over closed path

	$\oint \overline{H}.d\overline{L} = \oint_{1-2-3-4-1} \overline{H}.d\overline{L} = \int_1^2 \overline{H}.d\overline{L} + \int_2^3 \overline{H}.d\overline{L} + \int_3^4 \overline{H}.d\overline{L} + \int_4^1 \overline{H}.d\overline{L}$ $\oint \overline{H}.d\overline{L} = \oint_{1-2-3-4-1} \overline{H}.d\overline{L} = 0 + \int_2^3 H_x dx + 0 + \int_4^1 H_x dx$ $\oint \overline{H}.d\overline{L} = H_x \int_2^3 dx + H_x \int_4^1 dx = H_x b + H_x b = 2H_x b$ Equate the integral of $\overline{H}.\overline{dL}$ to the enclosed current $\oint \overline{H}.d\overline{L} = 2H_x b = I_{enc} = K_y b$ $H_x = \frac{1}{2} K_y$ Hence $\overline{H} = \frac{1}{2} K_y \overline{a}_x$ for z > 0 $\overline{H} = -\frac{1}{2} K_y \overline{a}_x$ for z < 0 Finally, Magnetic Field Intensity $\overline{H}$ can be expressed as $\overline{H} = \frac{1}{2} \overline{K} X \overline{a}_N$ A/m Where $\overline{a}_N$ is normal unit vector to the current sheet (In our assumption $\overline{a}_N = \overline{a}_z$) $\overline{H} = \frac{1}{2} \overline{K} X \overline{a}_N = \frac{1}{2} K_y \overline{a}_y X \overline{a}_z = \frac{1}{2} K_y \overline{a}_x$
4	**Apply Ampere's Circuit law to determine magnetic field intensity at any point in free space due to finite line current.** ANS: Consider a long straight conductor carrying direct current I placed along z-axis. Consider an Amperian closed path, enclosing the conductor



Consider point P on the closed at which $\bar{H}$ is to be obtained. The radius of the path is 'r' and hence P is at a perpendicular distance 'r' from the conductor.

Consider Cylindrical Co-ordinate System, The differential length vector $\overline{dL}$ from the Amperian path at point P in Cylindrical Co-ordinate System is $d\bar{L} = rd\phi\bar{a}_\phi$.

As H is in tangential direction $\bar{H} = H_\phi\bar{a}_\phi$.

Find $\bar{H}\cdot\overline{dL}$

$$\bar{H}\cdot d\bar{L} = H_\phi\,\bar{a}_\phi\cdot r\,d\phi\,\bar{a}_\phi = H_\phi\,r\,d\phi$$

Take integration of $\bar{H}\cdot\overline{dL}$ over closed path and equate its value to the direct current I. According to Ampere's Circuital Law

$$\oint \bar{H}\cdot d\bar{L} = I$$

$$\therefore \quad \int_{\phi=0}^{2\pi} H_\phi\,r\,d\phi = I$$

$$\therefore \quad H_\phi\,r\int_{\phi=0}^{2\pi} d\phi = I$$

$$\therefore \quad H_\phi\,r(2\pi) = I$$

$$\therefore \quad H_\phi = \frac{I}{2\pi r}$$

Hence the Magnetic Field Intensity $\bar{H}$ at point P is,

$$\bar{H} = H_\phi\,\bar{a}_\phi = \frac{I}{2\pi r}\,\bar{a}_\phi \quad A/m$$

5 **Derive the Maxwell's equations for time varying fields both in integral and differential forms.**

Maxwell's Equations are nothing but a set four expressions derived from Ampere's circuital Law, Faraday's Law, Gauss's Law for electric field and Gauss's Law for magnetic field.

These four equations can be written in the following forms

- Integral Form
- Point or differential Form

Maxwell's Equation 1

According Ampere's circuital Law, the line integral of magnetic field intensity $\overline{H}$ around a closed path is equal to the current enclosed by the path.

$$\oint_L \overline{H}.d\overline{L} = I_{enclosed}$$

$$\oint_L \overline{H}.d\overline{L} = \int_s \overline{J}.d\overline{S}$$

For time varying fields the original Ampere's circuital Law expression is changed by adding displacement current density to the conduction current density

$$\oint_L \overline{H}.d\overline{L} = \int_s [\overline{J} + \frac{\partial \overline{D}}{\partial t}].d\overline{S} \qquad .....(1.a)$$

This is the integral form Maxwell's Equation derived from Ampere's circuital Law.

The total magneto motive force around any closed path is equal to the surface integral of the conduction and displacement current densities over the entire surface bounded by the same closed path.

Apply Stroke's theorem to L.H.S of eq.,(1.a)

$$\int_s (\nabla X \overline{H}).d\overline{S} = \int_s [\overline{J} + \frac{\partial \overline{D}}{\partial t}].d\overline{S}$$

Cancel surface integral both sides

$$\nabla X \overline{H} = \overline{J} + \frac{\partial \overline{D}}{\partial t} \qquad .....(1.b)$$

This is the point form or differential form of Maxwell's Equation derived from Ampere's circuital Law.

Maxwell's Equation 2

According to Faraday's Law, In a closed circuit (path) the induced electromotive force (e.m.f) is equal to rate of decrease of

magnetic flux w.r.t time

$$\oint_L \overline{E}.d\overline{L} = -\frac{d\phi}{dt}$$

$$\oint_L \overline{E}.d\overline{L} = -\int_s \frac{\partial \overline{B}}{\partial t}.d\overline{S} \qquad .....(2.a)$$

This is the integral form Maxwell's Equation derived from Faraday's Law.

Apply Stroke's theorem to L.H.S of eq.,(2.a)

$$\int_s (\nabla X \overline{E}).d\overline{S} = -\int_s \frac{\partial \overline{B}}{\partial t}.d\overline{S}$$

Cancel surface integral both sides

$$\nabla X \overline{E} = -\frac{\partial \overline{B}}{\partial t} \qquad .....(2.b)$$

This is the point form or differential form of Maxwell's Equation derived from Faraday's Law

Maxwell's Equation 3

For electric field, according to Gauss's Law, In a closed surface the amount of net flux passing through it is equal to total charge enclosed by it.

$$\oint_s \overline{D}.d\overline{S} = Q_{enclosed}$$

$$\oint_s \overline{D}.d\overline{S} = \int_{vol} \rho_v dv \qquad .....(3.a)$$

This is the integral form Maxwell's Equation derived from Gauss's Law for electric field.

Apply Divergence's theorem to L.H.S of eq.,(3.a)

$$\int_{vol} \nabla.\overline{D} dv = \int_{vol} \rho_v dv$$

Cancel volume integral both sides

$$\nabla.\overline{D} = \rho_v \qquad .....(3.b)$$

This is the point form or differential form of Maxwell's Equation derived from Gauss's Law for electric field.

Maxwell's Equation 4

For magnetic field, according to Gauss's Law, the closed surface integral of magnetic flux density is always zero due to non existence of mono pole.

$$\oint_S \overline{B}.d\overline{S} = 0 \qquad .....(4.a)$$

This is the integral form Maxwell's Equation derived from Gauss's Law for magnetic field.

Apply Divergence's theorem to L.H.S of eq.,(4.a)

$$\int_{vol} \nabla.\overline{B}dv = 0$$

But being finite volume dv $\neq 0$

$$\nabla.\overline{B} = 0 \qquad .....(4.b)$$

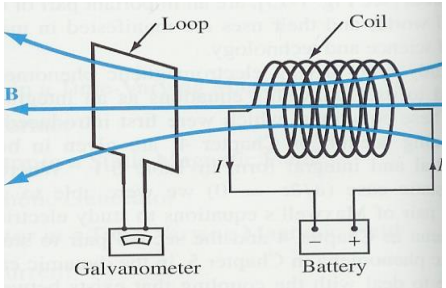
This is the point form or differential form of Maxwell's Equation derived from Gauss's Law for magnetic field.

Differential Form	Integral Form	Significance
$\nabla X\overline{H} = \overline{J} + \frac{\partial \overline{D}}{\partial t}$	$\oint_L \overline{H}.d\overline{L} = \int_s [\overline{J} + \frac{\partial \overline{D}}{\partial t}].d\overline{S}$	Ampere's Circuit Law
$\nabla X\overline{E} = -\frac{\partial \overline{B}}{\partial t}$	$\oint_L \overline{E}.d\overline{L} = -\int_s \frac{\partial \overline{B}}{\partial t}.d\overline{S}$	Faraday's Law
$\nabla.\overline{D} = \rho_v$	$\oint_s \overline{D}.d\overline{S} = \int_{vol} \rho_v dv$	Gauss's Law for Electric field
$\nabla.\overline{B} = 0$	$\oint_s \overline{B}.d\overline{S} = 0$	Gauss's Law for Magnetic field or non existence of mono pole

6	**What is the Inconsistency of Ampere's Law? How it is rectified by Maxwell?** For static EM fields, the Ampere's Law can be written as $\nabla X \overline{H} = \overline{J}$ .....(1) Take divergence both sides $\nabla.(\nabla X \overline{H}) = \nabla.\overline{J}$ But from the properties of Curl, 'the divergence of curl of any vector is zero' $\nabla.(\nabla X \overline{H}) = \nabla.\overline{J} = 0$ But, from continuity equation $\nabla.\overline{J} = 0$ for steady currents, that means for static EM fields it is satisfying. But for time varying fields, $\nabla.\overline{J} = -\frac{\partial \rho_v}{\partial t}$, not existing here. It is Inconsistency of Ampere's Law or it indicates $\nabla X \overline{H} = \overline{J}$ is incompatible for Time varying fields. The modification of Ampere's Law can perform just by adding one unknown term $\overline{N}$ for eq., (1) $\nabla X \overline{H} = \overline{J} + \overline{N}$ .....(2) Again take divergence both sides $\nabla.(\nabla X \overline{H}) = \nabla.\overline{J} + \nabla.\overline{N}$ Again from the properties of Curl $\nabla.(\nabla X \overline{H}) = \nabla.\overline{J} + \nabla.\overline{N} = 0$ $\nabla.\overline{J} = -\nabla.\overline{N}$ But for time varying fields $\nabla.\overline{J} = -\frac{\partial \rho_v}{\partial t} = -\nabla.\overline{N}$ According Gauss's Law $\nabla.\overline{N} = \frac{\partial \rho_v}{\partial t}$

	$$\nabla.\overline{D} = \rho_v$$ Thus replacing ρv by $\nabla.\overline{D}$ $$\nabla.\overline{N} = \frac{\partial(\nabla.\overline{D})}{\partial t}$$ $$\nabla.\overline{N} = \nabla.\frac{\partial \overline{D}}{\partial t}$$ Comparing both sides $$\overline{N} = \frac{\partial \overline{D}}{\partial t}$$ Now eq.,(2) can be written as $$\nabla X \overline{H} = \overline{J} + \frac{\partial \overline{D}}{\partial t}$$ $$\nabla X \overline{H} = \overline{J}_c + \overline{J}_D \qquad .....(3)$$ In eq.,(3) $\overline{J}_c$ = Conduction Current Density $\overline{J}_D$ = Displacement Current Density Now eq.,(3) is the point form or differential form of Maxwell's Equation derived from Ampere's Law. If we represent the above Maxwell's Equation in integral form, it can be written as $$\oint_L \overline{H}.d\overline{L} = I_{enc} = I_C + I_D = \int_s [\overline{J} + \frac{\partial \overline{D}}{\partial t}].d\overline{S}$$
7	**Derive an expression for vector magnetic potential.** **The Vector Magnetic Potential ($\overline{\mathbf{A}}$)** The Vector Magnetic Potential measured in Wb/m.

	The Vector Magnetic Potential is defined using the curl property 'Divergence of a curl of a vector is always zero' $$\nabla \cdot (\nabla \times \bar{A}) = 0$$ But $$\nabla \cdot \bar{B} = 0$$ from Maxwell's second eq., $$\bar{B} = \nabla \times \bar{A}$$ Thus Curl of Vector Magnetic Potential is Flux density. Now $$\nabla \times \bar{H} = \bar{J}$$ $$\nabla \times \frac{\bar{B}}{\mu_0} = \bar{J}$$ $$\nabla \times \bar{B} = \mu_0 \bar{J}$$ $$\nabla \times \nabla \times \bar{A} = \mu_0 \bar{J}$$ Using vector identity or Curl property $$\nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A} = \mu_0 \bar{J}$$ $$\bar{J} = \frac{1}{\mu_0}[\nabla \times \nabla \times \bar{A}] = \frac{1}{\mu_0}[\nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A}]$$ Thus if Vector Magnetic Potential is known then the current density $\bar{J}$ can be obtained. For defining $\bar{A}$ the current density need not be zero.
8	**Derive the expression for the emf induced by Faraday's disc generator with sketches.** ANS: According to Faraday's experiment, a static magnetic field cannot produce any current flow. But with a time varying field, an electromotive force (e.m.f) induces which may drive a current in a closed path or circuit. This e.m.f is nothing but a voltage that induces from 1. Changing magnetic fields 2. Motion of conductors in a magnetic field Faraday discovered that "The induced e.m.f is equal to the time rate of change of magnetic flux linking with in the closed

	circuit.” Faraday’s Law can be stated as, $$e = -N\frac{d\phi}{dt} Volts$$ where N = number of turns in the closed circuit e = induced e.m.f $$e = -N\frac{d\phi}{dt} \text{ volts.}$$  Let us assume single turn circuit i.e., N = 1, then Faraday’s Law can be stated as, $$e = -\frac{d\phi}{dt} \text{ volts}$$ the minus sign in the above expression indicates that the direction of the induced e.m.f
9	State and Prove boundary conditions between dielectric - dielectric boundary interface. The conditions of the magnetic field existing at the boundary of the two media when the magnetic field passes from one medium to other are called boundary conditions for magnetic fields or simply magnetic boundary conditions. When we consider magnetic boundary conditions, the conditions of $\overline{\mathbf{B}}$ and $\overline{\mathbf{H}}$ are studied at the boundary. The boundary between the two different magnetic materials is considered. To study conditions of $\overline{\mathbf{B}}$ and $\overline{\mathbf{H}}$ at the boundary, both the vectors are resolved into two components ; a) Tangential to boundary and b) Normal (perpendicular) to boundary. • Consider a boundary between two isotropic, homogeneous linear materials with different permeabilities μ_1 and μ_2 as shown in the Fig.

8.8.1. To determine the boundary conditions, let us use the closed path and the Gaussian surface.

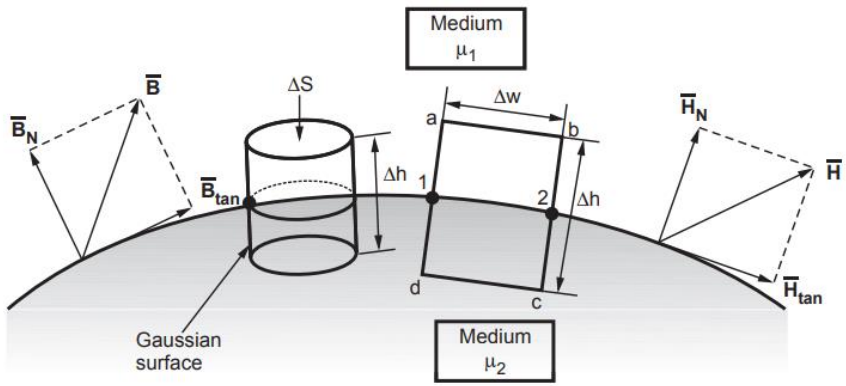


Fig. 8.8.1 Boundary between two magnetic materials of different permeabilities

1. Boundary Conditions for Normal Component

To find the normal component of $\overline{\mathbf{B}}$, select a closed Gaussian surface in the form of a right circular cylinder as shown in the Fig. 8.8.1. Let the height of the cylinder be Ah and be placed in such a way that Δh/2 is m medium 1 and remaining Δh/2 is in medium 2. Also the axis of the cylinder is in the normal direction to the surface.

• According to the Gauss's law for the magnetic field,

$$\oint_S \overline{\mathbf{B}} \bullet d\overline{\mathbf{S}} = 0 \quad \text{... (8.8.1)}$$

• The surface integral must be evaluated over three surfaces,

i) Top, ii) Bottom and iii) Lateral.

	• Let the area of the top and bottom is same, equal to ΔS. $$\therefore \oint_{top} \overline{\mathbf{B}} \bullet d\overline{\mathbf{S}} + \oint_{bottom} \overline{\mathbf{B}} \bullet d\overline{\mathbf{S}} + \oint_{lateral} \overline{\mathbf{B}} \bullet d\overline{\mathbf{S}} = 0 \quad \ldots (8.8.2)$$ • As we are very much interested in the boundary conditions, reduce Δh to zero. As Δh → 0, the cylinder tends to boundary and only top and bottom surfaces contribute in the surface integral. Thus surface integrals are calculated for top and bottom surfaces only. These surfaces are very small. Let the magnitude of normal component of $\overline{\mathbf{B}}$ be B_{N1} and B_{N2} in medium 1 and medium 2 respectively. As both the surfaces are very small, we can assume B_{N1} and B_{N2} constant over their surfaces. Hence we can write, For top surfaces

$$\oint_{Top} \overline{\mathbf{B}} \bullet d\overline{\mathbf{S}} = B_{N_1} \oint_{Top} d\overline{\mathbf{S}} = B_{N_1} \Delta S \qquad \text{... (8.8.3)}$$

For bottom surface

$$\oint_{Bottom} \overline{\mathbf{B}} \bullet d\overline{\mathbf{S}} = B_{N_2} \oint_{Bottom} d\overline{\mathbf{S}} = B_{N_2} \Delta S \qquad \text{... (8.8.4)}$$

For lateral surface

$$\oint_{Lateral} \overline{\mathbf{B}} \bullet d\overline{\mathbf{S}} = 0 \qquad \text{... (8.8.5)}$$

Putting values of surface integrals in equation (8.8.2), we get

$$B_{N_1} \Delta S - B_{N_2} \Delta S = 0 \qquad \text{... (8.8.6)}$$

From equation (8.8.6), we can write,

$B_{N1} \Delta S = B_{N2} \Delta S$

i.e. $B_{N1} = B_{N2}$ ... (8.8.7)

• Thus the normal component of $\overline{\mathbf{B}}$ is continuous at the boundary.

• As the magnetic flux density and the magnetic field intensity are related by,

$$\overline{\mathbf{B}} = \mu\overline{\mathbf{H}}$$

• Thus equation (8.8.7) can be written as,

$$\mu_1 H_{N_1} = \mu_2 H_{N_2}$$

$$\therefore \quad \boxed{\frac{H_{N_1}}{H_{N_2}} = \frac{\mu_2}{\mu_1} = \frac{\mu_{r2}}{\mu_{r1}}} \qquad \text{... (8.8.8)}$$

• Hence the normal component of $\overline{\mathbf{H}}$ is not continuous at the boundary. The field strengths in two media are inversely proportional to their relative permeabilities.

2. Boundary Conditions for Tangential Component

• According to Ampere's circuital law,

$$\oint \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}} = I \qquad \text{... (8.8.9)}$$

• Consider a rectangular closed path abcda as shown in the Fig. 8.8.1. It is traced in clockwise direction as a-b-c-d-a. This closed path is placed in a plane normal to the boundary surface. Hence $\oint \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}}$ can be divided into 6 parts.

$$\oint \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}} = \int_a^b \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}} + \int_b^1 \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}} + \int_1^c \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}} + \int_c^d \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}}$$

$$+ \int_d^2 \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}} + \int_2^a \overline{\mathbf{H}} \bullet d\overline{\mathbf{L}} = I \qquad \text{... (8.8.10)}$$

• From the Fig. 8.8.1 it is clear that, the closed path is placed in such a way that its two sides a-b and e-d are parallel to the tangential direction to the surface while the other two sides are normal to the surface at the boundary. This closed path is placed in such a way that half of its portion is in medium 1 and the remaining is in medium 2. The rectangular path is an elementary rectangular path with elementary height Ah and elementary width Δw. Thus over small width Δw, $\overline{\mathbf{H}}$ can be assume constant say H_{tan1} in medium 1 and H_{tan2} in medium 2. Similarly over a small height $\frac{\Delta h}{2}$, $\overline{\mathbf{H}}$ can be assumed constant say H_{N1} in medium 1 and H_{N2} in medium 2. Now assume that $\overline{\mathbf{K}}$ is the surface current normal to the path. Also from the Fig. 8.8.1 it is clear that the normal and tangential components in medium 1 and medium 2 are in opposite direction. Thus equation (8.8.10) can be written as,

$$K \bullet dw = H_{tan1}(\Delta w) + H_{N1}\left(\frac{\Delta h}{2}\right) + H_{N2}\left(\frac{\Delta h}{2}\right)$$

$$- H_{tan2}(\Delta w) - H_{N2}\left(\frac{\Delta h}{2}\right) - H_{N1}\left(\frac{\Delta h}{2}\right) \qquad \text{... (8.8.11)}$$

• To get conditions at boundary, $\Delta h \rightarrow 0$. Thus,

$K \cdot dw = H_{tan1}(\Delta w) - H_{tan2}(\Delta w)$

$H_{tan1} - H_{tan2}$

In vector form, we can express above relation by a cross product as

$$\boxed{\overline{H}_{tan1} - \overline{H}_{tan2} = \bar{a}_{N_{12}} \times \overline{K}} \qquad \ldots (8.8.13)$$

• where $\bar{a}_{N_{12}}$ is the unit vector in the direction normal at the boundary from medium 1 to medium 2.

• For $\overline{B}$, the tangential components can be related with permeabilities of two media using equation (8.8.12),

$$\therefore \frac{B_{tan1}}{\mu_1} - \frac{B_{tan2}}{\mu_2} = K \qquad \ldots (8.8.14)$$

• Consider a special case that the boundary is free of current. In other words, media are not conductors; so K = 0. Then equation (8.8.12) becomes

$H_{tan1} - H_{tan2} = 0$

Or $H_{tan1} - H_{tan2}$ ….. (8.8.15)

• For tangential components of $\overline{B}$ we can write,

$$\frac{B_{tan1}}{\mu_1} - \frac{B_{tan2}}{\mu_2} = 0$$

$$\therefore \quad \frac{B_{tan1}}{\mu_1} = \frac{B_{tan2}}{\mu_2}$$

$$\therefore \quad \boxed{\frac{B_{tan1}}{B_{tan2}} = \frac{\mu_1}{\mu_2} = \frac{\mu_{r1}}{\mu_{r2}}} \qquad \text{... (8.8.16)}$$

• From equations (8.8.15) and (8.8.16) it is clear that tangential component of $\overline{\mathbf{H}}$ are continuous, while tangential component of $\overline{\mathbf{B}}$ are discontinuous at the boundary, with the condition that the boundary is current free.

Let the fields make angles a_1 and a_2 with the normal to the interface as shown in the Fig. 8.8.2.

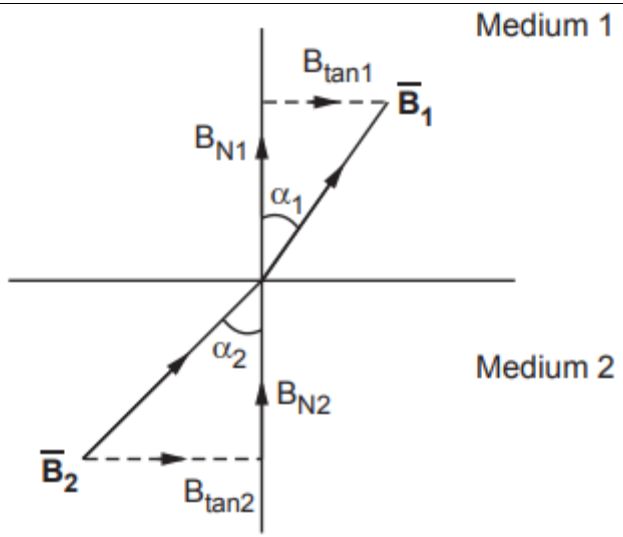


Fig. 8.8.2 Component of $\overline{B}$ at boundary

• Interms of angle a_1 and a_2, we can write relationship between normal components and tangential components of $\overline{B}$.

In medium 1,

$\tan \alpha 1 = B_{tan1} / B_{N1}$ ….. (8.8.17)

• Similarly in medium 2,

$\tan \alpha 2 = B_{tan2} / B_{N2}$ ….. (8.8.17)

As we know, $B_{N1} = B_{N2}$

• Consider an interface between air (medium 1) and soft iron (medium 2). For air, $\mu_{r1} = 1$. For soft iron, let $\mu_{r2} = 7000$. Then

$B_{tan1} / B_{tan2} = \tan \alpha 1 / \tan \alpha 2 = 1 / 7000$

	If α2 = 85°, then α1 = 0.093° and B_{tan1} = 0. • Thus practically when fields cross a medium of high μr to low μr, then the magnetic fields $\overline{B}$ and $\overline{H}$ are always perpendicular to the boundary.
10	A circular loop having radius 0.1 m and resistances 5Ω lies in the z = 0 plane. The circular loop is subjected to a magnetic field of $\overline{B}$ = 0.2 sin($10^3 t$) $\overline{a}_z$ (Wb/m^2). Find the current in the loop due to induced e.m.f. ANS: Given data Circular loop radius r = 0.1m Resistance R = 5Ω Magnetic flux density B = 0.2sin(10^3t)a_z wb/m^2 Total Flux (ϕ) $\Phi = \int_S \overline{B} \cdot d\overline{S}$ Induced e.m.f (e) $e = -\frac{d\phi}{dt}$ Current (i) $i = \frac{e}{R}$

MODEL

Total flux is given by

$$\Phi = \int_S \bar{B} \cdot d\bar{S}$$

$$= \int_{\Phi=0}^{2\pi} \int_{r=0}^{0.15} \left[(0.5 \sin 10^3 t)\bar{a}_z\right] \cdot \left[(r\,dr\,d\phi)\bar{a}_z\right]$$

$$= (0.5 \sin 10^3 t)[\phi]_0^{2\pi} \left[\frac{r^2}{2}\right]_0^{0.15}$$

$$= (0.5 \sin 10^3 t)\,[2\pi]\left[\frac{(0.15)^2}{2}\right]$$

$$= 35.3429 \sin 10^3 t \quad \text{mWb}$$

Now induced e.m.f. is given by,

$$e = -\frac{d\phi}{dt}$$

$$= -\frac{d}{dt}\left[35.3429 \times 10^{-3} \sin 10^3 t\right]$$

$$= -(35.3429 \times 10^{-3})(10^3)\cos 10^3 t$$

$$= -35.3429 \cos 10^3 t \quad \text{V}$$

Hence current in the conductor is given by,

$$i = \frac{e}{R} = \frac{-35.3429 \cos 10^3 t}{20}$$

$$\therefore \quad i = -1.7671 \cos 10^3 t \quad \text{A}$$

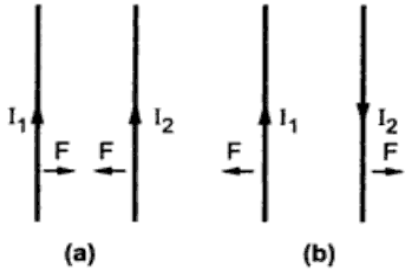
11

Derive an expression for Force between two straight long parallel current carrying conductors.

ANS:

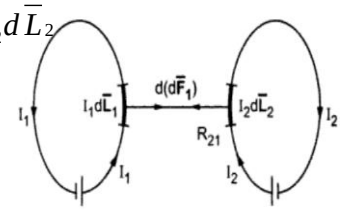
Consider two current carrying conductors are placed parallel to each other. Each of this conductor produces its own flux around it. When

two conductors are Placed close to each other, there exists a force due to the interaction of two fluxes. The force between two parallel current carrying conductors depends on the direction of the two currents.



a) If the directions of two currents are same, then the conductors experience a force of attraction
b) If the directions of two currents are opposite each other, then the conductors experience a force of repulsion.

Let us consider two current elements $I_1 d\overline{L}_1$ and $I_2 d\overline{L}_2$



Here the directions of I_1 and I_2 are same

Both the current elements produce their own magnetic fields . As the currents are flowing in the same direction force between these two current elements is the force of attraction.

If we want to take the force on current element1 due to current element2, that will be considered as $d(d\overline{F}_1)$ and it is noted as The force on current element1 ($I_1 d\overline{L}_1$) due to the magnetic field $d\overline{B}_2$ produced by current element2 ($I_2 d\overline{L}_2$)

$$d(d\overline{F}_1) = I_1 d\overline{L}_1 X d\overline{B}_2$$

According to Biot-Savart's Law the magnetic field produced by current element2 $I_2 d\overline{L}_2$ is given by, for free space.

$$d\overline{B}_2 = \mu_0 d\overline{H}_2 = \mu_0 \left[\frac{I_2 d\overline{L}_2 X \overline{a}_{R_{21}}}{4\pi R_{21}^2} \right]$$

Substituting the value of $d\overline{B}_2$ in $d(d\overline{F}_1)$

$$d(d\overline{F}_1) = \mu_0 \frac{I_1 d\overline{L}_1 X (I_2 d\overline{L}_2 X \overline{a}_{R_{21}})}{4\pi R_{21}^2}$$

By integrating $d(d\overline{F}_1)$ twice, the total force $\overline{F}_1$ on current element1 due to current element2 is given by

$$\overline{F}_1 = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{L_1} \oint_{L_2} \frac{d\overline{L}_1 \times \left(d\overline{L}_2 \times \overline{a}_{R21}\right)}{R_{21}^2}$$

Similarly, the total force $\overline{F}_2$ on current element2 due to current element1 is given by

$$\overline{F}_2 = \frac{\mu_0 I_2 I_1}{4\pi} \oint_{L_2} \oint_{L_1} \frac{d\overline{L}_2 \times \left(d\overline{L}_1 \times \overline{a}_{R12}\right)}{R_{12}^2}$$

$$\overline{F}_1 = -\overline{F}_2$$

For the two current carrying conductors of length *l* each, the magnitude of force exerted is given by

$$F = \frac{\mu I_1 I_2 l}{2\pi d}$$

Where

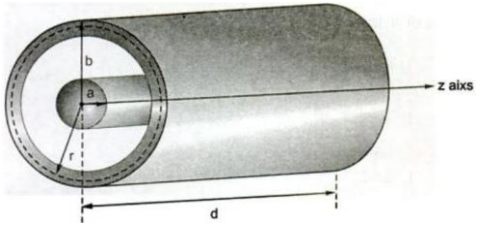
I_1 and I_2 are the currents through conductr1 and conductor2

d is the distance of separation between the two conductors

12

Derive an expression for inductance per unit length of long co-axial cable using Biot-Savart law.

ANS:



Consider a co-axial cable with Inner radius ‘a’ , Outer radius ‘b’

Let, The current through the co-axial cable as ‘I’, The length of cable is ‘d’

For the co-axial cable the field intensity between inner and outer conductors is given by

$$H = \frac{I}{2\pi r}$$

where a < r < b

If the axis of the co-axial cable is z-axis, the magnetic flux density with r = a to r = b and z = 0 to z = d is written as

$$\overline{B} = \frac{\mu I}{2\pi r}\overline{a}_{\varphi} T$$

The total magnetic flux is given by

$$\phi = \int_s \overline{B}.d\overline{S}$$

$$\phi = \frac{\mu_0 I d}{2\pi}\ln[\frac{b}{a}] \text{ webers}$$

Thus the inductance of a co-axial cable is given by

$$L = \frac{\text{Total flux linkage}}{\text{Total current}}$$

$$L = \frac{\phi}{I} \qquad L = \frac{\frac{\mu_0 I d}{2\pi}\ln[\frac{b}{a}]}{I} \qquad L = \frac{\mu_0 d}{2\pi}\ln[\frac{b}{a}]H$$

Thus the inductance of a co-axial cable is can also be expressed as per unit length

$$\frac{L}{d} = \frac{\mu_0}{2\pi} ln[\frac{b}{a}]H/m$$